# S.F. for "Output Feedback Stochastic Model Predictive Control for Discrete-time Stochastic Linear System via Block Coordinate Descent"

## Appendix A
## Matrix Definition

$$\mathcal{A}=\begin{bmatrix} I \\ A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times n_x},$$

$$\mathcal{K}=\begin{bmatrix} K_0 & 0 & \cdots & \cdots & 0 \\ 0 & K_1 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & K_{N-1} & 0 \\ 0 & \cdots & 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_u \times (N+1)n_x}$$

$$\mathcal{B}=\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ B & 0 & \cdots & 0 & 0 \\ AB & B & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 \\ A^{N-1}B & A^{N-2}B & \cdots & B & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_u},$$

$$\mathcal{D}=\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ D & 0 & \cdots & 0 & 0 \\ AD & D & \ddots & \vdots & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ A^{N-1}D & A^{N-2}D & \cdots & D & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_w},$$

$$\mathcal{C}=\begin{bmatrix} C & 0 & \cdots & 0 \\ 0 & C & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & C \end{bmatrix} \in \mathbb{R}^{(N+1)n_y \times (N+1)n_x}.$$

$$\mathcal{L}=\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ L_0 & 0 & \cdots & 0 & 0 \\ AL_0 & L_1 & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 \\ A^{N-1}L_0 & A^{N-2}L_1 & \cdots & L_{N-1} & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_y}.$$

Furthermore, we introduce the following augmented state vectors:

$$\begin{aligned} X &= \begin{bmatrix} x_0^T & x_1^T & x_2^T & \cdots & x_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ \widehat{X} &= \begin{bmatrix} \widehat{x}_0^T & \widehat{x}_1^T & \widehat{x}_2^T & \cdots & \widehat{x}_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ \overline{X} &= \begin{bmatrix} \overline{x}_0^T & \overline{x}_1^T & \overline{x}_2^T & \cdots & \overline{x}_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ U &= \begin{bmatrix} u_0^T & u_1^T & \cdots & u_{N-1}^T & 1_{n_u}^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_u}, \\ W_{(\bullet)} &= \begin{bmatrix} w_{(\bullet)0}^T & w_{(\bullet)1}^T & \cdots & w_{(\bullet)N-1}^T & 1_{n_w}^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_w}, \\ V &= \begin{bmatrix} v_0^T & v_1^T & \cdots & v_{N-1}^T & 1_{n_u}^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_u}, \end{aligned}$$

$$\begin{aligned} E &= \begin{bmatrix} e_0^T & e_1^T & \cdots & e_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ F &= \begin{bmatrix} f_0^T & f_1^T & \cdots & f_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}. \end{aligned}$$

$$\mathcal{V}=\left\{ V \;\middle|\; \begin{array}{l} \overline{X}=\mathcal{A}\overline{x}_0+\mathcal{B}V \\ \Sigma_E=\left[\mathbf{I}-\widetilde{\mathcal{L}}_y^k\mathcal{C} \;\; \widetilde{\mathcal{L}}_y^k\right]\begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2}\end{bmatrix}\begin{bmatrix}\left(\mathbf{I}-\widetilde{\mathcal{L}}_y^k\mathcal{C}\right)^T \\ \left(\widetilde{\mathcal{L}}_y^k\right)^T\end{bmatrix} \\ \Sigma_F=\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)\left[\mathbf{I} \quad \mathbf{I}\right]\begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)^T \\ a_{x,j}^T M_{t,x}\overline{X}+\Psi_{1-\Delta_{x,j}}^{-1} \\ \qquad \times\sqrt{a_{x,j}^T M_{t,x} 2\left(\Sigma_E+\Sigma_F\right)\left(a_{x,j}^T M_{t,x}\right)^T}\le b_{x,j} \\ a_{u,l}^T M_{t,u}V+\Psi_{1-\Delta_{u,l}}^{-1} \\ \qquad \times\sqrt{a_{u,l}^T M_{t,u}\Sigma_{F,K}\left(a_{u,l}^T M_{t,u}\right)^T}\le b_{u,l} \end{array} \right\}$$

$$\mathcal{Y}=\left\{ \widetilde{\mathcal{L}}_y \;\middle|\; \begin{array}{l} \overline{X}=\mathcal{A}\overline{x}_0+\mathcal{B}V^{k+1} \\ \Sigma_E=\left[\mathbf{I}-\widetilde{\mathcal{L}}_y\mathcal{C} \;\; \widetilde{\mathcal{L}}_y\right]\begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2}\end{bmatrix}\left[\mathbf{I}-\widetilde{\mathcal{L}}_y\mathcal{C} \;\; \widetilde{\mathcal{L}}_y\right]^T \\ \Sigma_F=\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)\left[\mathbf{I} \quad \mathbf{I}\right]\begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)^T \\ a_{x,j}^T M_{t,x}\overline{X}+\Psi_{1-\Delta_{x,j}}^{-1} \\ \qquad \times\sqrt{a_{x,j}^T M_{t,x} 2\left(\Sigma_E+\Sigma_F\right)\left(a_{x,j}^T M_{t,x}\right)^T}\le b_{x,j} \\ a_{u,l}^T M_{t,u}V^{k+1}+\Psi_{1-\Delta_{u,l}}^{-1} \\ \qquad \times\sqrt{a_{u,l}^T M_{t,u}\Sigma_{F,K}\left(a_{u,l}^T M_{t,u}\right)^T}\le b_{u,l} \\ M_{N,x} 2\left(\Sigma_E+\Sigma_F\right) M_{N,x}^T \preccurlyeq \Sigma_{x,N} \end{array} \right\}$$

$$\mathcal{Z}=\left\{ \widetilde{\mathcal{K}} \;\middle|\; \begin{array}{l} \overline{X}=\mathcal{A}\overline{x}_0+\mathcal{B}V^{k+1} \\ \Sigma_E=\left[\mathbf{I}-\widetilde{\mathcal{L}}_y^{k+1}\mathcal{C} \;\; \widetilde{\mathcal{L}}_y^{k+1}\right]\begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2}\end{bmatrix}\begin{bmatrix}\left(\mathbf{I}-\widetilde{\mathcal{L}}_y^{k+1}\mathcal{C}\right)^T \\ \left(\widetilde{\mathcal{L}}_y^{k+1}\right)^T\end{bmatrix} \\ \Sigma_F=\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}\right)\left[\mathbf{I} \quad \mathbf{I}\right]\begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}\right)^T \\ a_{x,j}^T M_{t,x}\overline{X}+\Psi_{1-\Delta_{x,j}}^{-1} \\ \qquad \times\sqrt{a_{x,j}^T M_{t,x} 2\left(\Sigma_E+\Sigma_F\right)\left(a_{x,j}^T M_{t,x}\right)^T}\le b_{x,j} \\ a_{u,l}^T M_{t,u}V^{k+1}+\Psi_{1-\Delta_{u,l}}^{-1} \\ \qquad \times\sqrt{a_{u,l}^T M_{t,u}\Sigma_{F,K}\left(a_{u,l}^T M_{t,u}\right)^T}\le b_{u,l} \\ M_{N,x} 2\left(\Sigma_E+\Sigma_F\right) M_{N,x}^T \preccurlyeq \Sigma_{x,N} \end{array} \right\}$$

for $t=0,\ldots,N-1$, $j=1,\ldots,m_x$ and $l=1,\ldots,m_u$.

## Appendix B
## Proof of Theorem 1

Before we proof the Theorem 1, we first provide the following results.

**Lemma 2**: Let $X\sim\mathcal{N}(\mu_X,\Sigma_X)$, $E\sim\mathcal{N}(0,\Sigma_E)$ and $F\sim\mathcal{N}(0,\Sigma_F)$ be random variables on $\mathbb{R}$. If they satisfy (37) and (11), then $\mu_X=\overline{X}$ and $\Sigma_X\preccurlyeq 2(\Sigma_E+\Sigma_F)$.

**Proof:** Since $\mu_X = \overline{X}$ is obvious, its proof is omitted.

Define $\Phi = \begin{bmatrix} E^T & F^T \end{bmatrix}^T \in \mathbb{R}^{2(N+1)n_x}$. Since $X = E + F + \overline{X}$, we have

$$\Sigma_X = [\mathbf{I}, \mathbf{I}]\, \Sigma_\Phi\, [\mathbf{I}, \mathbf{I}]^T$$

Furthermore, we have

$$\Sigma_\Phi = \begin{bmatrix} \Sigma_E & \mathrm{cov}(E,F) \\ \mathrm{cov}(E,F)^T & \Sigma_F \end{bmatrix}$$

Since for all $M \succcurlyeq 0$ and all $\begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \neq 0$, we have

$$\begin{aligned} &\begin{bmatrix} -z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \begin{bmatrix} -z_1 \\ z_2 \end{bmatrix} \\ &= z_1^T M_{11} z_1 + z_2^T M_{22} z_2 \\ &\quad - z_1^T M_{12} z_2 - z_2^T M_{12}^T z_1 \succcurlyeq 0, \end{aligned}$$

Thus, we can obtain that

$$\begin{aligned} &\begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \\ &= z_1^T M_{11} z_1 + z_2^T M_{22} z_2 + z_1^T M_{12} z_2 + z_2^T M_{12}^T z_1 \\ &\leq 2 z_1^T M_{11} z_1 + 2 z_2^T M_{22} z_2 \\ &= \begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} 2M_{11} & 0 \\ 0 & 2M_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \end{aligned}$$

Since $\Sigma_\Phi \succcurlyeq 0$, we have

$$\begin{bmatrix} \Sigma_E & \mathrm{cov}(E,F) \\ \mathrm{cov}(E,F)^T & \Sigma_F \end{bmatrix} \preceq \begin{bmatrix} 2\Sigma_E & 0 \\ 0 & 2\Sigma_F \end{bmatrix}$$

This implies $\Sigma_X = [\mathbf{I}, \mathbf{I}]\, \Sigma_\Phi\, [\mathbf{I}, \mathbf{I}]^T \preccurlyeq 2\Sigma_E + 2\Sigma_F$. This completes the proof. ■

Based on Lemma 2, we have

$$\begin{aligned} M_{0,x}\mu_X &= M_{0,x}\overline{X} \\ M_{0,x}\Sigma_X M_{0,x}^T &\preceq M_{0,x}\left(2\left(\Sigma_E + \Sigma_F\right)\right) M_{0,x}^T \end{aligned}$$

and

$$\begin{aligned} M_{N,x}\mu_X &= M_{N,x}\overline{X} \\ M_{N,x}\Sigma_X M_{N,x}^T &\preceq M_{N,x}\left(2\left(\Sigma_E + \Sigma_F\right)\right) M_{N,x}^T \end{aligned}$$

Therefore, we can derive the results directly.

## Appendix C
## Proof of Lemma 1

*Proof.* The objective function $\widetilde{J}(V, \widetilde{\mathcal{L}}_y, \widetilde{\mathcal{K}})$ is continuously differentiable and bounded below by zero (since $\mathcal{Q} \succcurlyeq 0$ and $\mathcal{R} \succcurlyeq 0$). In Algorithm 1, the sub-problems corresponding to $V$, $\widetilde{\mathcal{L}}_y$, and $\widetilde{\mathcal{K}}$ are rigorously converted into a strongly convex Quadratic Program (QP), a strongly convex Second-Order Cone Program (SOCP), and another strongly convex SOCP, respectively. Thus, each sub-problem is solved to global optimality for its respective block while fixing the other variables. With the proximal regularization terms $(\sigma/2)\|\cdot\|_F^2$ ($\sigma > 0$), we obtain the sufficient decrease chain:

$$\begin{aligned} &\widetilde{J}(V^{k+1}, \widetilde{\mathcal{L}}_y^{k+1}, \widetilde{\mathcal{K}}^{k+1}) + \frac{\sigma}{2}\left\|\Theta^{k+1} - \Theta^k\right\|_F^2 \\ &\leq \widetilde{J}(V^{k+1}, \widetilde{\mathcal{L}}_y^{k+1}, \widetilde{\mathcal{K}}^k) \leq \widetilde{J}(V^{k+1}, \widetilde{\mathcal{L}}_y^k, \widetilde{\mathcal{K}}^k) \leq \widetilde{J}(V^k, \widetilde{\mathcal{L}}_y^k, \widetilde{\mathcal{K}}^k) \end{aligned}$$

Since the sequence of objective values is non-increasing and bounded below, it converges to a finite infimum and $\lim_{k\to\infty} \|\Theta^{k+1} - \Theta^k\|_F = 0$. Under the strong convexity of the individual QP and SOCP sub-problems, the minimizer for each block is uniquely attained. According to the convergence theory for proximal block coordinate descent methods on semi-algebraic second-order cone programs satisfying the Kurdyka–Łojasiewicz property [33, 57], the entire sequence $\{\Theta^k\}$ has finite length and converges globally to a stationary point. This completes the proof. □

## Appendix D
## Proof of Theorem 2

*Proof.* Assume that at time $t-1$, Problem 3 is feasible and yields the converged stationary sequence $V_{t-1}^*$, $\widetilde{\mathcal{L}}_{y,t-1}^*$, and $\widetilde{\mathcal{K}}_{t-1}^*$ (as guaranteed by Lemma 1). At time $t$, the controller selects the initial condition according to (80).

**Case 1:** Problem 3 is feasible with the current estimation $(\hat{\mu}_{x,t}, \hat{\Sigma}_{x,t})$. The initialization strategy directly accepts it, i.e., $(\mu_{x,0|t}, \Sigma_{x,0|t}) = (\hat{\mu}_{x,t}, \hat{\Sigma}_{x,t})$. A feasible solution explicitly exists, rendering the problem trivially feasible.

**Case 2:** Problem 3 is infeasible with the current estimation. The strategy falls back to the deterministic surrogate bounds optimized from the previous step: $(\mu_{x,0|t}, \Sigma_{x,0|t}) = \left(\overline{x}_{1|t-1}^*, 2\left(\Sigma_{E,1|t-1}^* + \Sigma_{F,1|t-1}^*\right)\right)$. We construct a candidate shifted sequence by appending the terminal gains $K_f$, $L_f$, and a nominal terminal input $v_f = 0$:

$$\begin{aligned} V_t^{cand} &= [v_{1|t-1}^*, \ldots, v_{N-1|t-1}^*, 0] \\ K_t^{cand} &= [K_{1|t-1}^*, \ldots, K_{N-1|t-1}^*, K_f] \\ L_t^{cand} &= [L_{1|t-1}^*, \ldots, L_{N-1|t-1}^*, L_f] \end{aligned}$$

By initializing strictly with the surrogate upper bound optimized at $t-1$, the nominal states and surrogate covariance matrices for the first $N-1$ steps perfectly match the optimal tail of the previous iteration. Specifically, for all $i \in \{0, \ldots, N-1\}$, we have the exact index-shifted propagation:

$$\overline{x}_{i|t}^{cand} = \overline{x}_{i+1|t-1}^*, \Sigma_{E,i|t}^{cand} = \Sigma_{E,i+1|t-1}^*, \Sigma_{F,i|t}^{cand} = \Sigma_{F,i+1|t-1}^*$$

Substituting these shifted variables into the tightened surrogate state chance constraints at time $t$ yields:

$$\begin{aligned} &a_{x,j}^T \overline{x}_{i|t}^{cand} + \Psi_{1-\Delta_x}^{-1}\sqrt{2a_{x,j}^T\left(\Sigma_{E,i|t}^{cand} + \Sigma_{F,i|t}^{cand}\right)a_{x,j}} \\ &= a_{x,j}^T \overline{x}_{i+1|t-1}^* + \\ &\Psi_{1-\Delta_x}^{-1}\sqrt{2a_{x,j}^T\left(\Sigma_{E,i+1|t-1}^* + \Sigma_{F,i+1|t-1}^*\right)a_{x,j}} \\ &\leq b_{x,j} \end{aligned}$$

which are strictly satisfied since the optimal sequence at $t-1$ was strictly feasible. By the property established in Lemma 2 ($\Sigma_X \preccurlyeq 2\left(\Sigma_E + \Sigma_F\right)$), satisfying these surrogate constraints rigorously guarantees the satisfaction of the actual physical chance constraints:

$$a_{x,j}^T \mu_{X,i|t} + \Psi_{1-\Delta_x}^{-1}\sqrt{a_{x,j}^T \Sigma_{X,i|t} a_{x,j}} \leq b_{x,j}$$

Similar index-shifted inequality propagation holds for the input chance constraints and the terminal covariance constraint

$M_N(\Sigma_X) M_N^T \preccurlyeq \Sigma_{x,N}$. For the final prediction step $N$, the shifted terminal state enters the terminal invariant set $\Omega_f$. Under Assumption 1, the local laws ensure all probabilistic terminal constraints and covariance bounds are met. Therefore, the constructed shifted sequences constitute a strictly feasible solution. By mathematical induction, the system maintains recursive feasibility. □

## APPENDIX E
## PROOF OF THEOREM 3

*Proof.* Since $\mathcal{Q} \geq 0$ and $\mathcal{R} \geq 0$, we have that $l^m_{i|t} \geq 0$. Then $J^m(\mu_{X,t}, V_t)$ is positive. From the recursive feasibility established previously, $v_{i|t} \in \mathcal{U}$ is bounded and so is $\overline{x}_{i|t} \in \mathcal{X}$. Then $J^m(\mu_{X,t}, V_t)$ is bounded.

When the system utilizes Strategy 2, the initialization strictly follows the prediction, and we have the standard deterministic decrease:

$$\begin{aligned} &J^m(\mu_{X,t}, V_t) - J^m(\mu_{X,t-1}, V_{t-1}) \\ &\leq v_{N|t}^T R v_{N|t} + \overline{x}_{N+1|t-1}^{*T} Q \overline{x}_{N+1|t-1}^* \\ &\quad - v_{0|t-1}^{*T} R v_{0|t-1}^* - \overline{x}_{0|t}^{*T} Q \overline{x}_{0|t}^* \leq 0 \end{aligned}$$

When Strategy 1 is triggered, the initial nominal state incorporates the latest measurement estimate $\hat{\mu}_{x,t}$. The deviation between the measurement estimate and the prior prediction $\overline{x}^*_{1|t-1}$ is governed by the bounded estimation error dynamics characterized by $\Sigma_E$. Because the objective function is locally Lipschitz continuous within the compact feasible sets $\mathcal{X}$ and $\mathcal{U}$, there exists a finite positive constant $C_w$ bounded by the noise covariances $\Sigma_{w_1}$ and $\Sigma_{w_2}$ such that the cost variation induced by the measurement update is bounded:

$$J^m(\hat{\mu}_{X,t}, V_t) \leq J^m(\overline{x}^*_{1|t-1}, V_{t-1}) + C_w$$

Therefore, the expected value of the Lyapunov function strictly decreases outside a bounded set defined by $C_w$, establishing that the nominal closed-loop system is practically stable and converges to a bounded region around the origin rather than exactly to zero. □

## APPENDIX F
## PROOF OF THEOREM 4

*Proof.* Let the optimal cost at time $t$ be $J_t^* = J(\Sigma^*_{X,t}, \mu^*_{X,t}, K_t^*, V_t^*)$. At time $t+1$, under recursive feasibility, the shifted sequence $(V_{t+1}^{cand}, K_{t+1}^{cand}, L_{t+1}^{cand})$ with $v_f = 0$ and terminal gains $K_f, L_f$ constitutes a valid feasible solution. Let the cost associated with this candidate sequence be $J_{t+1}^{cand}$. By optimality, we have:

$$J^*_{t+1} \leq J^{cand}_{t+1} \tag{1}$$

The candidate cost $J^{cand}_{t+1}$ at time $t+1$ is constructed by removing the first stage cost of time $t$ and adding the terminal stage cost. Before addressing the closed-loop measurement update, we first explicitly derive the algebraic upper bound for the terminal stage cost $l^{cand}_{N+1|t} = l(\overline{x}^*_{N+1|t}, v_f, \Sigma^*_{x_{N+1|t}}, K_f)$.

From the cost definition, the terminal stage cost consists of nominal and variance components:

$$\begin{aligned} l^{cand}_{N+1|t} &= \overline{x}^{*T}_{N+1|t} Q \overline{x}^*_{N+1|t} + v_f^T R v_f \\ &\quad + \mathrm{Tr}\left(\Sigma^*_{x_{N+1|t}} Q\right) + \mathrm{Tr}\left(K_f \Sigma^*_{F,N+1|t} K_f^T R\right) \end{aligned}$$

Since $v_f \in \mathcal{U}$ and $\overline{x}^*_{N+1|t} \in \mathcal{X}$ within the terminal invariant set, their nominal costs are bounded by finite maximums: $v_f^T R v_f \leq J_v^{max}$ and $\overline{x}^{*T}_{N+1|t} Q \overline{x}^*_{N+1|t} \leq J_x^{max}$.

From the surrogate terminal constraint and Lemma 2, we have $\Sigma^*_{x_{N+1|t}} \preccurlyeq 2(\Sigma_E + \Sigma_F) \preccurlyeq \Sigma_{x,N}$. Thus, $\mathrm{Tr}\left(\Sigma^*_{x_{N+1|t}} Q\right) \leq \mathrm{Tr}(\Sigma_{x,N} Q)$. From the input chance constraint reformulation over the compact polytopic input set $\mathcal{U} = \{u \in \mathbb{R}^{n_u} \mid a_{u,l}^T u \leq b_{u,l},\, l = 1, \ldots, m_u\}$ whose normal matrix $A_u = [a_{u,1}, \ldots, a_{u,m_u}]^T \in \mathbb{R}^{m_u \times n_u}$ has full column rank ($\lambda_{\min}(A_u^T A_u) > 0$), we know:

$$\begin{aligned} &\Psi^{-1}_{1-\Delta_{u,l}} \sqrt{a_{u,l}^T K_f \Sigma^*_{F,N+1|t} K_f^T a_{u,l}} \leq b_{u,l}, \quad l = 1, \ldots, m_u \\ &\implies \mathrm{Tr}\Big(K_f \Sigma^*_{F,N+1|t} K_f^T R\Big) \leq \frac{\lambda_{\max}(R)}{\lambda_{\min}(A_u^T A_u)} \sum_{l=1}^{m_u} \left( \frac{b_{u,l}}{\Psi^{-1}_{1-\Delta_{u,l}}} \right)^2 \end{aligned}$$

Defining the deterministic terminal upper bound $\xi$ as:

$$\begin{aligned} \xi &= J_v^{max} + J_x^{max} + \mathrm{Tr}(\Sigma_{x,N} Q) \\ &\quad + \frac{\lambda_{\max}(R)}{\lambda_{\min}(A_u^T A_u)} \sum_{l=1}^{m_u} \left( \frac{b_{u,l}}{\Psi^{-1}_{1-\Delta_{u,l}}} \right)^2 \end{aligned} \tag{2}$$

we algebraically guarantee that $l^{cand}_{N+1|t} \leq \xi$.

Unlike a nominal deterministic case, the actual realization of the candidate cost at $t+1$ depends on the closed-loop initialization strategy. If Strategy 1 is triggered, the state undergoes a measurement update, introducing an innovation jump defined as $\delta_t = \hat{x}_t - \overline{x}^*_{1|t-1}$.

Since the actual control input applied is $u_{t-1} = v^*_{0|t-1} + K^*_{0|t-1}(\hat{x}_{t-1} - \overline{x}^*_{0|t-1})$, and Strategy 1 implies $\overline{x}^*_{0|t-1} = \hat{x}_{t-1}$, the feedback term vanishes, yielding $u_{t-1} = v^*_{0|t-1}$. Substituting the measurement model $y_{t-1} = C x_{t-1} + w_{2,t-1}$ into the observer dynamics, the nominal terms perfectly cancel out, simplifying the innovation jump to:

$$\delta_t = L^*_{t-1}(C e_{0|t-1} + w_{2,t-1}) \tag{3}$$

Given that $e_{0|t-1}$ and $w_{2,t-1}$ are zero-mean and mutually independent, it follows that $\mathbb{E}[\delta_t] = 0$. The covariance of this jump, denoted as $\Sigma_\delta$, is strictly determined by the observer gain and noise statistics:

$$\Sigma_\delta = \mathbb{E}[\delta_t \delta_t^T] = L^*_{t-1}\left(C \Sigma^*_{E,0|t-1} C^T + \Sigma_{w_2}\right) L^{*T}_{t-1} \tag{4}$$

As the candidate control sequence $V^{cand}_{t+1}$ is open-loop shifted, this zero-mean jump $\delta_t$ propagates linearly through the prediction horizon as $A^i \delta_t$. Evaluating its expectation within the nominal quadratic cost $J^m$, the cross-terms vanish ($\mathbb{E}[\delta_t] = 0$), yielding an explicit closed-form bound $\Gamma$ for the expected additional innovation cost:

$$\Gamma = \sum_{i=0}^{N} \mathrm{Tr}\left((A^i)^T Q A^i \Sigma_\delta\right) \tag{5}$$

This explicit mathematical formulation is of paramount importance: it not only proves that $\Gamma$ is a finite bound (restricted by noise variances and the prediction horizon), but also rigorously justifies the necessity of the proposed BCD algorithm. By

jointly optimizing the observer gain $L_t$ iteratively, the algorithm structurally minimizes the quadratic term $\Sigma_\delta$, thereby systematically suppressing the innovation cost $\Gamma$.

Taking the conditional expectation given the information at time $t$, the expected cost difference is upper bounded by the terminal cost addition ($\xi$), the initial stage cost reduction, and the explicit innovation bound $\Gamma$:

$$\begin{aligned}\mathbb{E}_w[J_{t+1}^{cand}] - J_t^* &\le l_{N+1|t}^{cand} - l(x_{0|t}^*, u_{0|t}^*) + \Gamma \\ &\le \xi + \Gamma - l(x_{0|t}^*, u_{0|t}^*) \end{aligned} \tag{6}$$

Defining the unified closed-loop bound $\overline{\xi} = \xi + \Gamma$, and utilizing the optimality condition $J_{t+1}^* \le J_{t+1}^{cand}$, we obtain:

$$\mathbb{E}_w[J_{t+1}^*] - J_t^* \le \overline{\xi} - l(x_{0|t}^*, u_{0|t}^*) \tag{7}$$

Taking the total expectation and summing this inequality from $t = 0$ to $T$ yields a telescoping sum without requiring the open-loop assumption $v_{i+1|t}^* = v_{i|t+1}^*$:

$$\sum_{t=0}^{T} \mathbb{E}_w(l(x_{0|t}^*, u_{0|t}^*)) \le J_0^* - \mathbb{E}_w(J_{T+1}^*) + (T+1)\overline{\xi} \tag{8}$$

Since $J_{T+1}^* \ge 0$, we drop the $-\mathbb{E}_w(J_{T+1}^*)$ term. Dividing both sides by $T$ and taking the limit as $T \to \infty$, we obtain:

$$\lim_{T\to\infty} \frac{1}{T} \mathbb{E}_w \left( \sum_{t=0}^{T} l(x_{0|t}^*, u_{0|t}^*) \right) \le \overline{\xi} \tag{9}$$

This rigorously proves that the expected average stage cost is strictly bounded for the closed-loop stochastic system, explicitly capturing both terminal constraint bounds and measurement feedback innovations. □

## APPENDIX G
## SUPPLEMENTARY SIMULATION RESULTS AND EXTENDED ANALYSIS

### *A. Offline Terminal Ingredients and 300-Run Initialization Sensitivity*

For the numerical example in Section V-A with open-loop unstable plant matrices

$$A = \begin{bmatrix} 1.02 & 0.95 \\ 0.06 & 1.02 \end{bmatrix}, \quad B = \begin{bmatrix} 0.85 \\ 1.00 \end{bmatrix}, \quad D = I_2, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

solving the discrete algebraic Riccati equations and discrete Lyapunov equation offline for $(A, B, Q, R)$ and $(A^T, C^T, D\Sigma_{w_1}D^T, \Sigma_{w_2})$ with $Q = I_2$, $R = 0.1$, $\Sigma_{w_1} = \mathrm{diag}(0.008, 0.012)$, and $\Sigma_{w_2} = 0.008$ yields the stabilizing terminal gains, steady-state covariance bound, and invariant ellipsoid required by Assumption 1, Remark 10, and terminal constraints (53)–(55):

$$\begin{aligned} K_f &= \begin{bmatrix} -0.6365 & -1.0317 \end{bmatrix}, \quad L_f = \begin{bmatrix} 1.3477 \\ 0.5752 \end{bmatrix}, \\ \rho(A + BK_f) &= 0.3679 < 1, \quad \rho(A - L_f C) = 0.3940 < 1, \\ \Sigma_{x,\text{bound}} &= 2\left(\Sigma_{e,\infty} + \Sigma_{f,\infty}\right) = \begin{bmatrix} 0.3260 & 0.0636 \\ 0.0636 & 0.1842 \end{bmatrix}, \\ \Sigma_{x,N} &= 0.85\, M_{N,x} 2\left(\Sigma_{E,\text{ss}} + \Sigma_{F,\text{ss}}\right) M_{N,x}^T, \\ P_f &= \begin{bmatrix} 1.7398 & 0.1280 \\ 0.1280 & 1.1157 \end{bmatrix}, \\ \Omega_f &= \left\{ \overline{x} \in \mathbb{R}^2 : \overline{x}^T P_f \overline{x} \le 2.798 \right\}, \end{aligned}$$

with tightened terminal bounds $b_{x,\text{tight}} = 1.6388$ and $b_{u,\text{tight}} = 1.7620$. Table I summarizes the sensitivity of Algorithm 1 to initial covariance scaling $(\Sigma_E^0, \Sigma_F^0) = cI$ across $c \in \{0.1, 0.5, 1.5, 3.0\}$ and across 300 initializations spanning $c \in [0.05, 2.0]$. Across all scaling regimes, the BCD algorithm converges in 5 outer iterations to $J_0^* = 19.3885$ in Case 1 and to $J_0^* = 23.7853 \pm 0.0039$ with a relative standard deviation of $0.016\%$ in Case 2.

TABLE I
SENSITIVITY OF BCD CONVERGENCE TO INITIAL COVARIANCE SCALING $(\Sigma_E^0, \Sigma_F^0) = cI$ $(N = 8)$

| Initialization Scale $(\Sigma_E^0, \Sigma_F^0) = cI$ | Case 1 / 2 BCD Iters | Case 1 Initial $J_0^0$ | Case 1 Final $J_0^*$ | Case 2 Initial $J_0^0$ | Case 2 Final $J_0^*$ |
|---|---|---|---|---|---|
| $c = 0.10$ | 5 / 5 | 185.82 | 19.3885 | 100.30 | 23.7848 |
| $c = 0.50$ (Default) | 5 / 5 | 211.50 | 19.3885 | 125.98 | 23.7874 |
| $c = 1.50$ | 5 / 5 | 275.70 | 19.3885 | 190.18 | 23.7843 |
| $c = 3.00$ | 5 / 5 | 372.00 | 19.3885 | 286.48 | 23.7799 |
| 300 Runs ($c \in [0.05, 2.0]$) | 5 / 5 | — | 19.3885 | — | $23.7853 \pm 3.9 \times 10^{-3}$ |

### *B. Quantitative Gain-Strategy Ablation & Step-by-Step Recursion Verification*

Table II provides the quantitative comparison of the four gain-design strategies depicted in Figs. 1 and 2 of the main paper over 100 Common-Random-Number Monte Carlo realizations. Here, the constant gains $(L_{\text{fix}}, K_{\text{fix}}) = (0.30L_{\text{ss}}, 0.22K_{\text{lqr}})$ are nominal stabilizing gains with Schur spectral radii $\rho(A + BK_{\text{fix}}) = 0.8804$ and $\rho(A - L_{\text{fix}}C) = 0.8573$, which ensures that $2\Sigma_{E,N}(L_{\text{fix}}) \prec \Sigma_{x,N}$ so all four ablation strategies remain strictly feasible under the offline terminal covariance bounds (53)–(55). Furthermore, because the $(N+1)$-block prediction matrices retain the initial time step $t = 0$ with dimensions $p_{\text{full}} = (N+1)n_x$, $p_{y,\text{full}} = (N+1)n_y$, and $\mu = Nn_u$, they map bijectively to the causal step-by-step innovation and disturbance-feedback gains. Consequently, the block-matrix propagation in (8a)–(8f) and the step-by-step causal state-space recursion yield identical trajectories and terminal covariances up to machine precision below $10^{-14}$.

TABLE II
QUANTITATIVE ABLATION OF GAIN-DESIGN STRATEGIES OVER 100 COMMON-RANDOM-NUMBER TRIALS IN CASE 1 AND CASE 2 $(N = 8)$

| Case / Gain Strategy | Surrogate Cost $J_0$ | Empirical Cost $J_{\text{emp}} \pm \text{CI}_{95}$ | Terminal Cov. $\text{Tr}(\Sigma_{X,N}^{\text{emp}} / \Sigma_{X,N}^{\text{true}})$ | Tracking RMSE | State / Input Viol. (%) | Solve Time (ms) |
|---|---|---|---|---|---|---|
| **Case 1** ($\Delta_u = 0.10$) | | | | | | |
| Proposed (TV $L_t$, TV $K_t$) | **19.39** | **18.06 ± 0.38** | **0.308 / 0.287** | **0.421** | **0.0% / 0.0%** | 19.09 |
| TV $K_t$, Fixed $L_t$ | 20.84 | 18.76 ± 0.56 | 0.473 / 0.427 | 0.488 | 0.0% / 0.0% | 21.60 |
| TV $L_t$, Fixed $K_t$ | 24.66 | 18.86 ± 0.66 | 0.427 / 0.397 | 0.492 | 0.0% / 0.0% | 8.29 |
| Fixed $L_t$ and Fixed $K_t$ | 23.53 | 21.61 ± 1.53 | 1.713 / 1.618 | 0.679 | 0.0% / 0.0% | 0.25 |
| **Case 2** ($\Delta_x = \Delta_u = 0.20$) | | | | | | |
| Proposed (TV $L_t$, TV $K_t$) | **23.79** | **22.38 ± 0.41** | **0.284 / 0.287** | **0.400** | **4.0% / 0.0%** | 37.94 |
| TV $K_t$, Fixed $L_t$ | 25.25 | 22.89 ± 0.53 | 0.352 / 0.427 | 0.446 | 4.0% / 0.0% | 26.61 |
| TV $L_t$, Fixed $K_t$ | 29.33 | 23.11 ± 0.56 | 0.345 / 0.391 | 0.446 | 3.0% / 0.0% | 17.30 |
| Fixed $L_t$ and Fixed $K_t$ | 28.15 | 25.24 ± 1.21 | 1.235 / 1.618 | 0.604 | 3.0% / 0.0% | 3.90 |

### *C. Robustness Against Non-Gaussian Disturbances*

Table III evaluates the proposed BCD-SMPC in Case 2 ($N = 8$, 200 Monte Carlo trials) across four zero-mean, covariance-matched disturbance distributions: (i) Gaussian $\mathcal{N}(0, \Sigma_w)$ with $\Psi_{1-\Delta}^{-1} = 0.8416$, (ii) Uniform $\mathcal{U}[-\sqrt{3}\sigma, \sqrt{3}\sigma]$, (iii) heavy-tailed Laplace $\text{Laplace}(0, \sigma/\sqrt{2})$, and (iv) heavy-tailed Student-$t$ ($\nu = 5$) with Cantelli bound $\kappa_{\text{DR}} = 2.0$. In all

cases, the empirical violation rates remain strictly below the $20\%$ risk threshold.

TABLE III
CLOSED-LOOP ROBUSTNESS UNDER NON-GAUSSIAN DISTURBANCES IN CASE 2 ($N = 8$, $\Delta_x = \Delta_u = 20\%$)

| Disturbance Distribution | Tightening Factor $\kappa_x$ | Empirical Cost $J_{\text{emp}}$ | Terminal $\text{Tr}(\Sigma^{\text{emp}}_{X,N})$ | State Viol. Rate (%) | Input Viol. Rate (%) |
|---|---|---|---|---|---|
| Gaussian $\mathcal{N}(0, \Sigma_w)$ | Gaussian (0.8416) | 22.29 | 0.285 | 2.5% | 0.0% |
| Uniform $\mathcal{U}[-\sqrt{3}\sigma, \sqrt{3}\sigma]$ | Cantelli (2.0000) | 26.32 | 0.324 | 0.0% | 0.0% |
| Heavy-Tailed Laplace | Cantelli (2.0000) | 26.40 | 0.352 | 0.5% | 0.0% |
| Heavy-Tailed Student-$t$ ($\nu = 5$) | Cantelli (2.0000) | 26.21 | 0.259 | 0.0% | 0.0% |

### *D. Control Experiment With Unscaled Steady-State DARE Gains ($L_{base} = L_{ss}$, $K_{base} = K_{lqr}$)*

To verify that our conclusions do not rely on the nominal constant-gain scaling $(L_{\text{base}}, K_{\text{base}})$, Table IV reports a control experiment in which all six competing baselines use the unscaled infinite-horizon DARE steady-state Kalman gain $L_{\text{base}} = L_{\text{ss}}$ and LQR gain $K_{\text{base}} = K_{\text{lqr}}$. Even compared against unscaled infinite-horizon DARE gains, the proposed BCD-SMPC achieves competitive constraint-compliant empirical cost of 22.04 in Case 2 and the lowest cost of 27.98 on the 6-DOF vehicle, while reducing the terminal covariance trace on the 6-DOF vehicle by 2.27–2.72 times from 0.052–0.063 down to 0.023 with $0.0\%$ constraint violations.

TABLE IV
BASELINE CONTROL EXPERIMENT WITH UNSCALED INFINITE-HORIZON DARE GAINS ($L_{\text{BASE}} = L_{\text{SS}}$, $K_{\text{BASE}} = K_{\text{LQR}}$)

| Benchmark / Method | Empirical Cost $J_{\text{emp}} \pm \text{CI}_{95}$ | $\text{Tr}(\Sigma^{\text{emp}}_{X,N}/\Sigma^{\text{true}}_{X,N})$ | Tracking RMSE | State / Input Viol. (%) |
|---|---|---|---|---|
| **Case 2** ($N = 6$) | | | | |
| Proposed BCD-SMPC | **22.04 ± 0.43** | 0.301 / 0.313 | 0.409 | **3.0% / 0.0%** |
| CS-SMPC ($L_{\text{ss}}$) | 21.03 ± 0.35 | 0.232 / 0.251 | 0.401 | 19.0% / 0.0% |
| Tube-MPC ($L_{\text{ss}}$, $K_{\text{lqr}}$) | 21.44 ± 0.37 | 0.232 / 0.253 | 0.400 | 6.0% / 0.0% |
| RF-MPC ($L_{\text{ss}}$, $K_{\text{lqr}}$) | 22.13 ± 0.39 | 0.232 / 0.253 | 0.400 | 3.0% / 0.0% |
| DR-MPC ($L_{\text{ss}}$) | 22.31 ± 0.43 | 0.306 / 0.322 | 0.405 | 3.0% / 0.0% |
| DDRMPC ($L_{\text{ss}}$, $K_{\text{lqr}}$) | 22.43 ± 0.41 | 0.232 / 0.253 | 0.400 | 2.0% / 0.0% |
| SDP-MPC ($L_{\text{ss}}$) | 22.49 ± 0.40 | 0.231 / 0.253 | 0.403 | 2.0% / 0.0% |
| **Case 4 (6-DOF Vehicle,** $N = 12$) | | | | |
| Proposed BCD-SMPC | **27.98 ± 5.00** | **0.023 / 0.023** | **0.287 m** | **0.0% / 0.0%** |
| Tube-MPC ($L_{\text{ss}}$, $K_{\text{lqr}}$) | 32.04 ± 5.12 | 0.063 / 0.072 | 0.297 m | 0.0% / 0.0% |
| CS-SMPC ($L_{\text{ss}}$) | 32.40 ± 5.14 | 0.062 / 0.072 | 0.294 m | 0.0% / 3.0% |
| SDP-MPC ($L_{\text{ss}}$) | 48.85 ± 5.06 | 0.052 / 0.057 | 0.345 m | 0.0% / 4.0% |
| DDRMPC ($L_{\text{ss}}$, $K_{\text{lqr}}$) | 55.37 ± 5.65 | 0.063 / 0.072 | 0.297 m | 0.0% / 0.0% |
| RF-MPC ($L_{\text{ss}}$, $K_{\text{lqr}}$) | 102.05 ± 6.90 | 0.063 / 0.072 | 0.297 m | 0.0% / 0.0% |
| DR-MPC ($L_{\text{ss}}$) | 264.69 ± 15.02 | 0.173 / 0.196 | 0.342 m | 0.0% / 0.0% |

### *E. Complete 6-DOF AutoRally Racing Vehicle Simulation Protocol, Jacobian Matrices, and Fair Baseline Comparison*

To ensure complete transparency and reproducibility of Section V-C without any ambiguity, we detail the exact mathematical model, discretization, track geometry, constraint tightening, and baseline evaluation protocol implemented in `bcd_smpc.py` and `run_all.py`:

**1) State Vector, Equilibrium, and Exact ZOH Discretization:** The 6-DOF curvilinear dynamic bicycle tracking-error state is ordered as $\tilde{x} = [e_y,\, v_y,\, e_\psi,\, w_z,\, \tilde{v}_x,\, \tilde{s}]^T \in \mathbb{R}^6$ with control input $u = [\delta,\, a]^T \in \mathbb{R}^2$ and physical parameters listed in Table V. The front and rear static normal tire loads are $f_{Fz} = mg\ell_R/(\ell_F + \ell_R) = 87.09$ N and $f_{Rz} = mg\ell_F/(\ell_F + \ell_R) = 128.73$ N, giving peak Pacejka lateral forces $D_f = f_{Fz}\,\text{tire}_D = 95.80$ N and $D_r = f_{Rz}\,\text{tire}_D = 141.61$ N. Along the reference racing line with $v_x = v_g = 9.0$ m/s, $\rho(s) = \rho_0 + \rho' \tilde{s}$, $\rho_0 = 0.25\ \text{m}^{-1}$, $\rho' = 0.05\ \text{m}^{-2}$, and nominal steady-state yaw rate $w_{z,\text{ref}} = \rho_0 v_g$, the origin $\tilde{x}_{\text{eq}} = \mathbf{0}, u_{\text{eq}} = \mathbf{0}$ is an exact equilibrium satisfying $f_{\text{veh}}(\mathbf{0}, \mathbf{0}) = \mathbf{0}$ identically. Evaluating the continuous-time Jacobian matrices $A_c = \left.\frac{\partial f_{\text{veh}}}{\partial \tilde{x}}\right|_{(\mathbf{0},\mathbf{0})}$ and $B_c = \left.\frac{\partial f_{\text{veh}}}{\partial u}\right|_{(\mathbf{0},\mathbf{0})}$ gives:

$$A_c = \begin{bmatrix} 0 & 1 & 9.00 & 0 & 0 & 0 \\ 0 & -4.67 & 0 & -9.00 & 0 & 0 \\ -0.56 & 0 & 0 & 1 & -0.25 & -0.45 \\ 0 & 0 & 0 & -7.30 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 2.25 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 & 0 \\ 16.96 & 0 \\ 0 & 0 \\ 115.33 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The discrete-time matrices $A$ and $B$ with sampling period $\Delta t = 0.12$ s and horizon $N = 12$ are obtained via exact zero-order-hold matrix exponentiation:

$$\begin{bmatrix} A & B \\ \mathbf{0}_{2\times 6} & \mathbf{I}_2 \end{bmatrix} = \exp\left( \begin{bmatrix} A_c & B_c \\ \mathbf{0}_{2\times 6} & \mathbf{0}_{2\times 2} \end{bmatrix} \Delta t \right).$$

**2) Partial-Observation Model, Noise Statistics, and Polytopic Constraints:** Only lateral error $e_y$, heading error $e_\psi$, and longitudinal speed error $\tilde{v}_x$ are measured with $n_y = 3 < n_x = 6$, while lateral slip velocity $v_y$, yaw rate $w_z$, and arc-length drift $\tilde{s}$ are unmeasured:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad A_x = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad b_x = \begin{bmatrix} 0.65 \\ 1.80 \\ 1.20 \end{bmatrix},$$

and input box constraints $A_u = [\mathbf{I}_2,\, -\mathbf{I}_2]^T \in \mathbb{R}^{4\times 2}$, $b_u = [0.50,\, 2.50,\, 0.50,\, 2.50]^T$ with $\Delta_x = \Delta_u = 0.10$. Under uniform Bonferroni risk allocation, $\Delta_{x,j} = \Delta_x/3 = 0.0333$ gives $\Psi^{-1}_{1-\Delta_{x,j}} = 1.8339$ and $\Delta_{u,l} = \Delta_u/4 = 0.0250$ gives $\Psi^{-1}_{1-\Delta_{u,l}} = 1.9600$ across $t = 1, \ldots, N$.

**3) S-Bend Chicane Centerline Mapping and Closed-Loop Evaluation:** In Figs. 6–7, the S-bend dirt chicane centerline curvature profile $\kappa_{\text{ref}}(s) = 0.23\exp(-((s-6.5)/3.6)^2) - 0.23\exp(-((s-19.5)/3.6)^2)$ is integrated via cumulative trapezoidal quadrature to obtain the reference heading $\psi_{\text{ref}}(s)$ and Cartesian centerline $(X_{\text{ref}}(s), Y_{\text{ref}}(s))$ with unit normal vector $n(s) = [-\sin\psi_{\text{ref}}(s),\, \cos\psi_{\text{ref}}(s)]^T$. Both the nominal trajectory and each simulated sample trajectory $(e^{(s)}_{y,t}, \tilde{s}^{(s)}_t)$, together with the analytical $3\sigma$ position uncertainty ellipses with semi-axes $3\sqrt{\Sigma_{x,t}[1,1]}$ and $3\sqrt{\Sigma_{x,t}[6,6]}$, are mapped onto global track coordinates via $[X_t, Y_t]^T = [X_{\text{ref}}(s_{\text{node},t} + 2\tilde{s}_t), Y_{\text{ref}}(s_{\text{node},t} + 2\tilde{s}_t)]^T + 2e_{y,t}\, n(s_{\text{node},t} + 2\tilde{s}_t)$. All methods are evaluated over the same 100 Common-Random-Number trials with fixed random seed 400, both on the discrete-time linearized model and by integrating the continuous-time nonlinear Pacejka ODEs via 4th-order Runge–Kutta without drift subtraction.

**4) Fair Baseline Protocol and Physical Interpretation of Figs. 6(a)–(b):** All six competing baselines and the fixed-gain ablations share the Schur-stable observer gain $L_{\text{base}} = L_{\text{fix}} = 0.70 L_{\text{ss}}$ with $\rho(A - L_{\text{fix}}C) = 0.9671 < 1$ and nominal feedback gain $K_{\text{base}} = K_{\text{fix}} = 0.65 K_{\text{lqr}}$ with $\rho(A + BK_{\text{fix}}) = 0.8476 < 1$, while Table IV reports the control experiment with unscaled $(L_{\text{ss}}, K_{\text{lqr}})$. Because $n_y = 3 < n_x = 6$ leaves $[v_y, w_z, \tilde{s}]^T$ unmeasured, the steady-state observer mode with $\rho(A - L_{\text{ss}}C) = 0.9528$ contracts much more slowly than state feedback with $\rho(A + BK_{\text{lqr}}) = 0.8684$, and estimation error $\Sigma_E$ directly excites $\Sigma_F = (\mathbf{I} + \mathcal{B}\widetilde{\mathcal{K}})(z_f + \Sigma_E)(\mathbf{I} + \mathcal{B}\widetilde{\mathcal{K}})^T$. Consequently, optimizing time-varying $L_t$ in both Fig. 6(a) and

Fig. 6(b) reduces $\mathrm{Tr}(\Sigma_{E,N})$ from $0.0221$ to $0.0048$–$0.0057$, which explains why Figs. 6(a)–(b) achieve $\mathrm{Tr}(\Sigma^{\mathrm{emp}}_{X,N}) = 0.023$ and $0.026$ compared with $0.073$ and $0.083$ for the Fixed-$L_t$ strategies in Figs. 6(c)–(d). Jointly optimizing time-varying $K_t$ in Fig. 6(a) further reduces the velocity and yaw-rate tracking variances so that $\mathrm{Tr}(\Sigma_{F,N})$ drops from $0.0126$ to $0.0098$ and surrogate $\mathrm{Tr}(2(\Sigma_{E,N}+\Sigma_{F,N}))$ drops from $0.0367$ to $0.0293$, satisfying terminal constraint (55) with margin $+0.0548$ whereas TV $L_t$ Fixed $K_t$ violates (55) with margin $-0.8468$. In the warm-start benchmark, the proposed BCD-SMPC converges in 3 iterations and $61.46$ ms under cold start, in 2 iterations and $57.44$ ms under offline LQG warm start at $t = 0$, and in $2.0$ iterations and $28.44$ ms mean step time with zero surrogate constraint violations across receding-horizon steps $t = 1, \ldots, 15$.

TABLE V
PHYSICAL PARAMETERS OF THE 6-DOF AUTORALLY OFF-ROAD RACING VEHICLE MODEL

| Parameter | Symbol | Value |
|---|---|---|
| Vehicle mass / Yaw moment of inertia | $m$ / $I_z$ | 22.0 kg / 1.10 kg $\cdot$ m$^2$ |
| Front / Rear CG-to-axle distance | $\ell_F$ / $\ell_R$ | 0.34 m / 0.23 m |
| Front / Rear static normal tire load | $f_{Fz}$ / $f_{Rz}$ | 87.09 N / 128.73 N |
| Pacejka stiffness / shape / peak factors | $B_{f,r}$ / $C_{f,r}$ / tire$_D$ | 4.10 / 0.95 / 1.10 |
| Target longitudinal speed / Chicane curvature | $v_g$ / $\rho_0$ / $\rho'$ | 9.0 m/s / 0.25 m$^{-1}$ / 0.05 m$^{-2}$ |
| Sampling period / Prediction horizon | $\Delta t$ / $N$ | 0.12 s / 12 |
| State / Input / Measured output dimensions | $n_x$ / $n_u$ / $n_y$ | 6 / 2 / 3 |